Introduction to Low Resolution Spectrograph

Prism can be used as a dispersing element in a spectrograph based on a theory that the refractive index is different depending on the wavelength. However, the spectral resolution of a prism is much lower than of the diffraction grating. The prism spectrograph is the traditional form of spectrograph first introduced by Bunsen and Kirchhoff. In this experiment, we will use a prism low resolution spectrograph called **BESOS** or **BE**st **S**imple **O**ptical **S**pectrograph from the CAOS group. BESOS is an easy assembling spectrograph. It consists of two doublets used as collimator and doublet respectively. Prism is applied as a dispersive element. We will be able to build a simple spectrograph from scratch starting from the optical design, optomechanical assembly. And later we will use this spectrograph to observe some bright object on the sky. Further information is available in https://spectroscopy.wordpress.com/2011/04/10/besos-a-prism-spectrograph/

Surf	Type	Radius	Thickness	Glass	Diameter
OBJ	STANDARD	Infinity	27.68252		1
1	STANDARD	70.28889	1.5	SF10	12.5
2	STANDARD	15.50989	3.5	SK2	12.5
3	STANDARD	-19.668	10		12.5
4	COORDBRK	-	0		-
STO	STANDARD	Infinity	0	SF18	10
6	COORDBRK	-	5		-
7	COORDBRK	-	0		-
8	STANDARD	Infinity	0		10
9	COORDBRK	-	10		-
10	STANDARD	25.119	4	N-BK7	12.5
11	STANDARD	-17.783	2.3	N-SF5	12.5
12	STANDARD	-49.403	37.03728		12.5
IMA	STANDARD	Infinity			7.41799

Figure 1: ZEMAX Lens data for BESOS

Lens data in Figure 1 can be plugged in ZEMAX and check the performance of the spectrograph through the analysis tool in ZEMAX e.g. spot diagram. Then we can assemble the optical elements and the optomechanical parts of the spectrograph according to Figure 2 and 3.

In order to observe spectra, we have to connect the spectrograph with a CCD. We can acquire spectra from bright sources such as LED, halogen lamp or the sun. Proper calibration image sets are needed. Later on, the data reduction and wavelength calibration is required.



Figure 2: BESOS assembly

REFERENCES

J. F. James, *Spectrograph Design Fundamental*, Cambridge Press, 2007

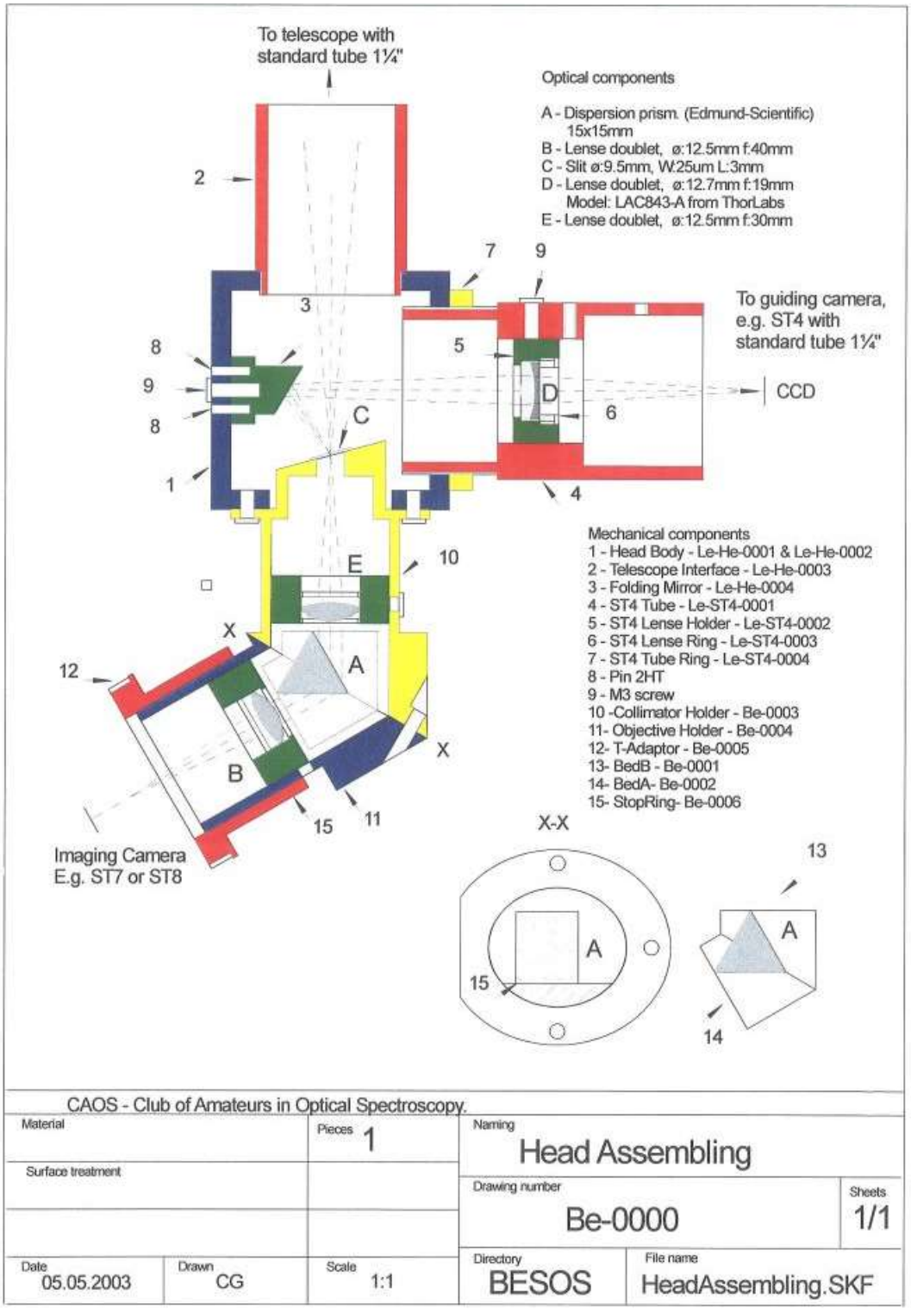


Figure 3: Optomechanical assembly plan for BESOS